MIMIC-III NLP Assignment

Comparing spaCy, scispaCy, and medspaCy on Real ICU Discharge Notes

AI 395T — AI for Healthcare | Summer 2026 | Murali Balarama Chakaravarthy

BACKGROUND

Why Compare Three NLP Tools on Clinical Text?

The Problem It Solves

General NLP tools miss a lot of clinical meaning - they don't know 'MI' means heart attack, or that 'denies pain' is a negative finding.

What Makes It Different

spaCy is general-purpose. scispaCy adds biomedical training. medspaCy adds clinical context - negation, family history, hypothetical statements.

Why This Matters

Choosing the right NLP tool changes what information you can reliably pull out of real patient notes.

Cohort: MIMIC-III discharge summaries, patients diagnosed with subarachnoid hemorrhage (ICD-9 430)

SETUP

How the Pipeline Is Structured

Cohort & Data Source

- Data pulled from Google BigQuery (MIMIC-III public dataset)
- Cohort: patients with ICD-9 code 430 (subarachnoid hemorrhage)
- Note type: Discharge summaries only
- 888 matching notes found; a smaller sample used for processing speed
- De-identification brackets and extra whitespace cleaned before NLP

What Each Tool Does

- Entity extraction: find clinical terms in the text
- word2vec: turn words into number vectors based on context
- t-SNE: compress those vectors down to 2D so we can plot and compare them
- Same steps run for spaCy, scispaCy, and medspaCy on the same notes

spaCy — General-Purpose NLP

The everyday baseline: fast, well-tested, not trained on medical text

What it does:

Runs standard named entity recognition (people, places, dates, numbers) plus word2vec and t-SNE on the same notes.

Why it matters:

It's a fair baseline to measure the other two tools against - free of any medical training.

How it was run:

Load en_core_web_sm, extract entities, tokenize, train word2vec, plot tuned t-SNE.

Finding

3,104 entities found across 17 label types, but the top ones were CARDINAL, ORG, and PERSON - mostly noise like numbers and section headers, not clinical concepts.

Word2vec vocabulary size: 2,306 words

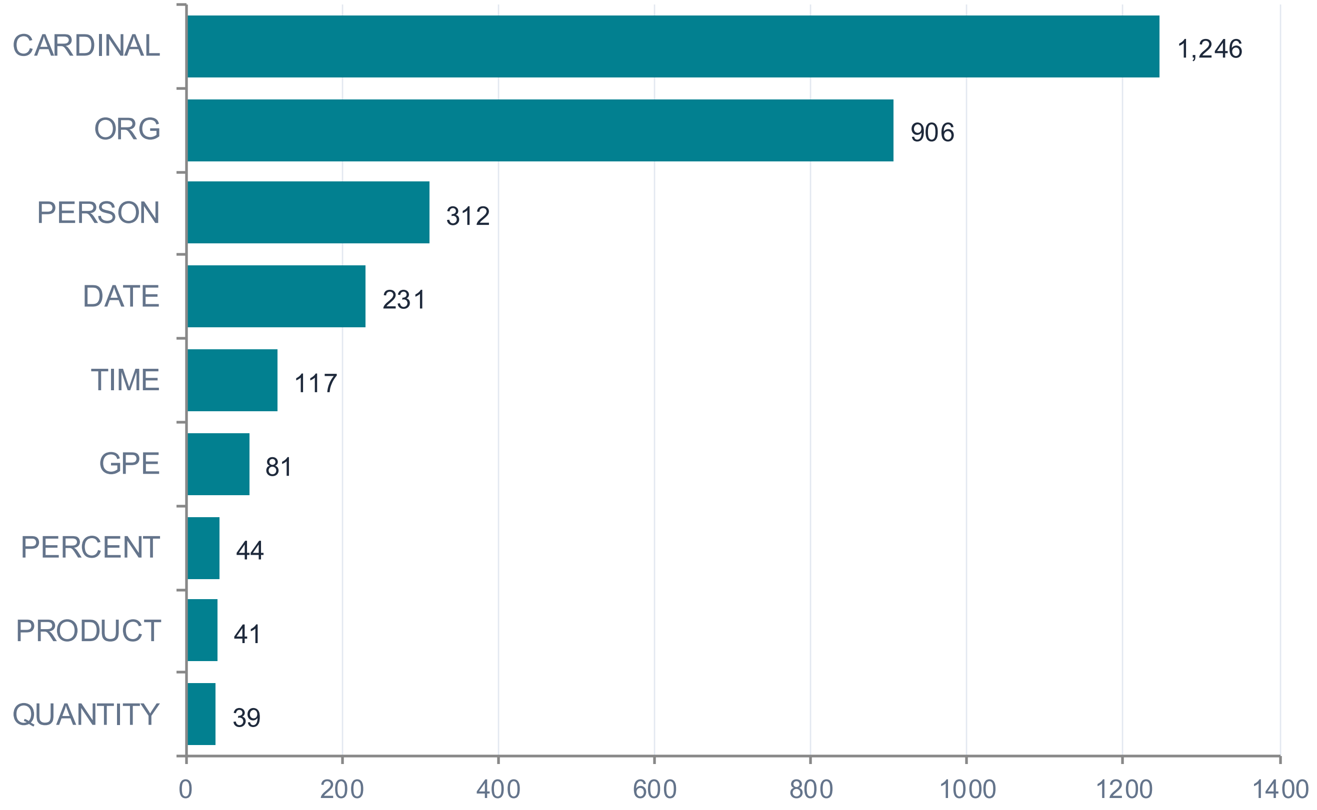
Model: en_core_web_sm | Output: entity counts, word2vec, tuned t-SNE plot

TOOL 1 OF 3

spaCy — Extracted Entities & word2vec

Extracted Entities (by label)

Examples: "Addendum" -> PERSON, "PO" -> ORG, "30" -> CARDINAL (mostly noise)



word2vec Embeddings

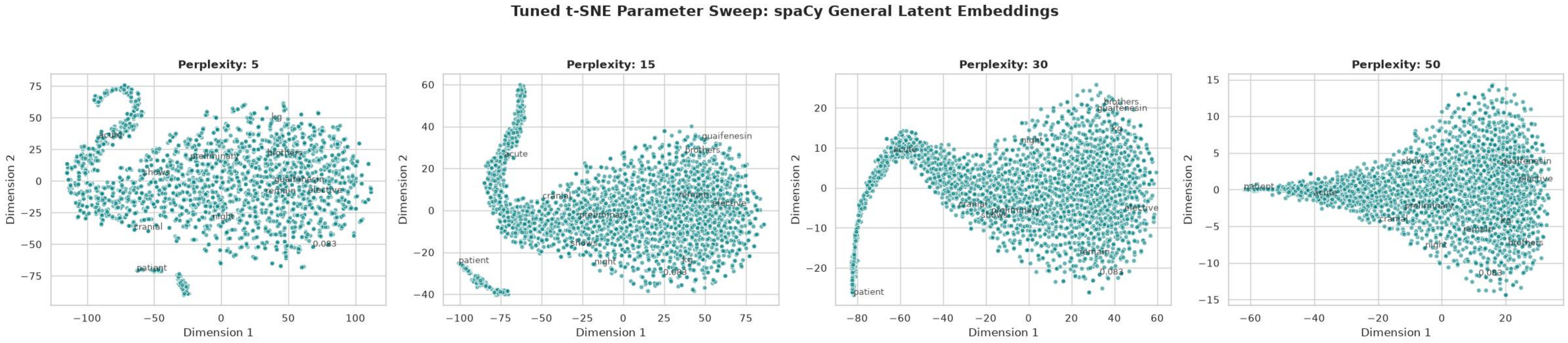
2,306

vocabulary words learned

- Vector size: 100 dimensions
- Window: 5 words of context
- Min count: 2 occurrences
- Trained on: tokenized note text from this tool's pipeline

spaCy — Tuned t-SNE Plot

word2vec embeddings projected to 2D across four perplexity values



Perplexity swept at 5, 15, 30, and 50 to check the clustering held up across settings, per the Distill.pub tuning guidance.

scispaCy — Biomedical NLP

Same pipeline, trained on biomedical and clinical text

What it does:

Uses models trained on biomedical literature to recognize diseases, chemicals, and other domain-specific terms.

Why it matters:

It understands clinical vocabulary that general spaCy simply doesn't recognize.

How it was run:

Load en_core_sci_sm and en_ner_bc5cdr_md, extract entities, tokenize, train word2vec, plot tuned t-SNE.

Finding

1,986 entities found - almost entirely DISEASE (1,207) and CHEMICAL (779) - a far more clinically useful breakdown than spaCy's 17 generic categories.

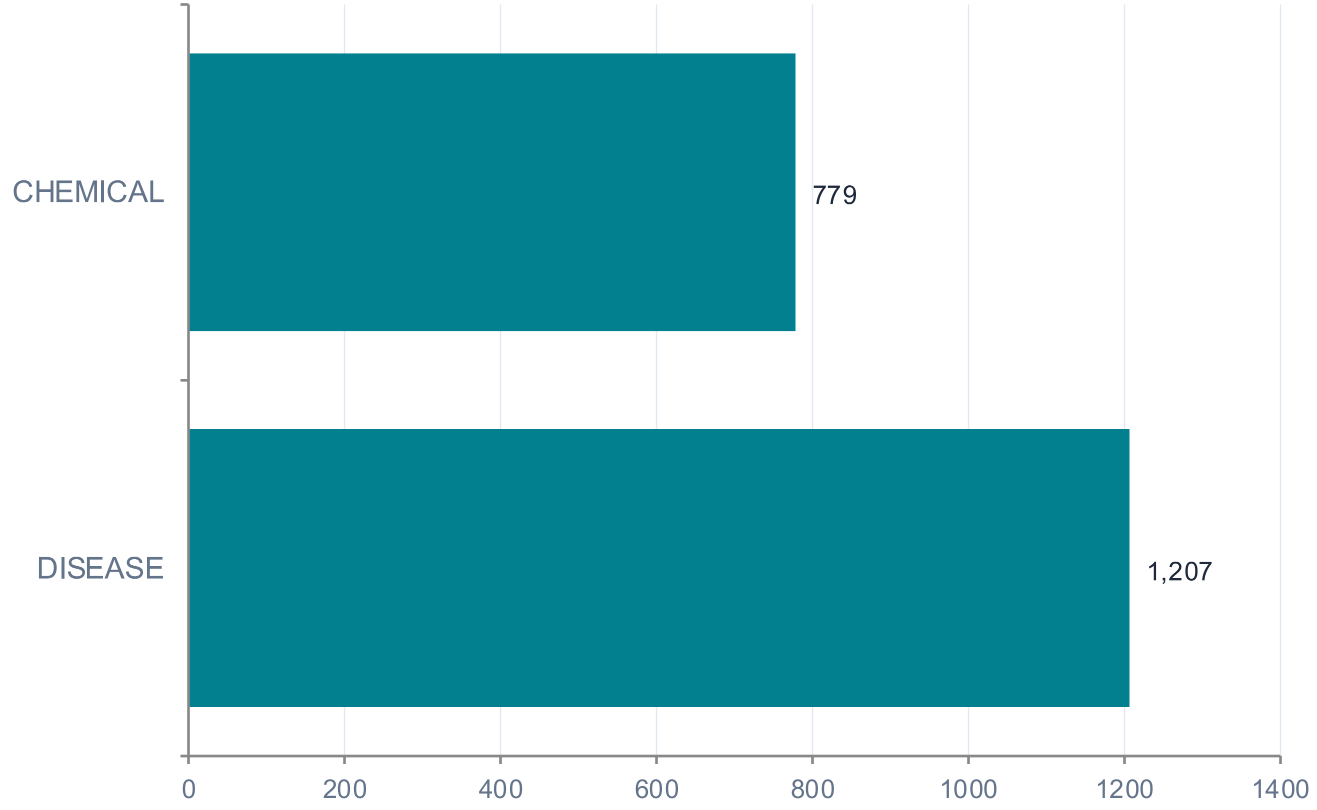
Word2vec vocabulary size: 2,455 words

Models: en_core_sci_sm, en_ner_bc5cdr_md | Output: entity counts, word2vec, tuned t-SNE plot

scispaCy — Extracted Entities & word2vec

Extracted Entities (by label)

Examples: "aneurysm" -> DISEASE, "nimodipine" -> CHEMICAL, "atrial fibrillation" -> DISEASE



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scispaCy — Tuned t-SNE Plot

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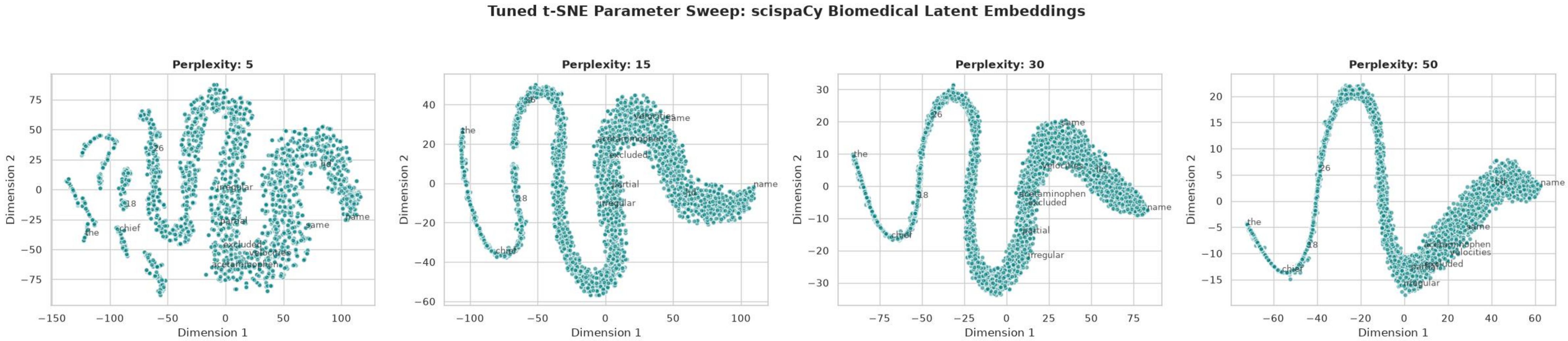


Biomedical vocabulary (disease and chemical terms) forms tighter, more separable clusters than spaCy's general-purpose embeddings.

medspaCy — Clinical Context Detection

Adds negation, family history, and hypothetical detection on top of entities

What it does:

Flags whether a clinical concept is negated, historical, hypothetical, or about a family member - not just whether it's mentioned.

Why it matters:

'Denies chest pain' and 'has chest pain' are opposite findings. Only medspaCy tells them apart.

How it was run:

Load medspaCy, add target rules for SAH-related concepts, run context detection over the sample.

Finding

254 entities found (needed 15 custom target rules first). Of these, 58 were flagged as negated - catching findings spaCy and scispaCy can't tell apart from a positive mention.

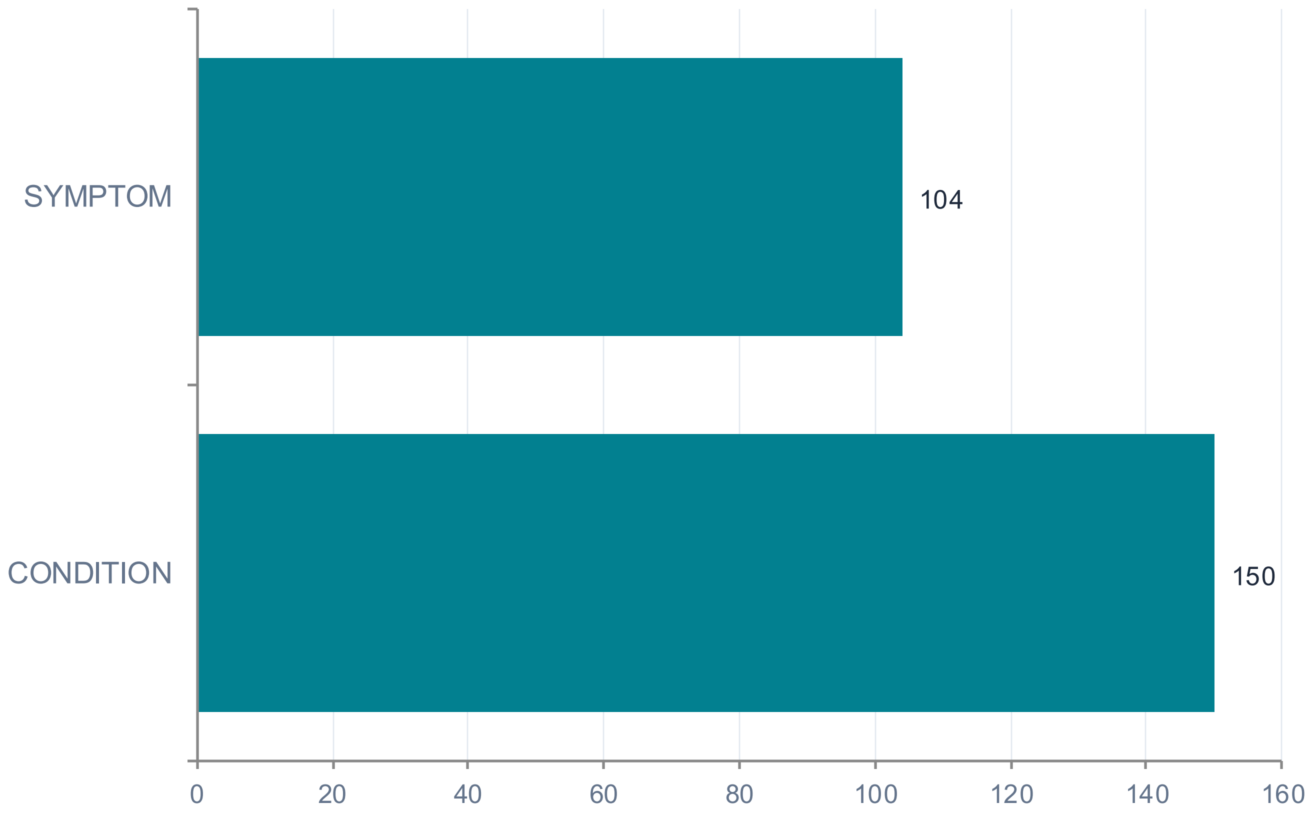
Word2vec vocabulary size: 2,386 words

Package: medspaCy | Output: entity + negation/context flags, word2vec, tuned t-SNE plot

medspaCy — Extracted Entities & word2vec

Extracted Entities (by label)

Examples: "hemorrhage" -> CONDITION, negated=True; "headache" -> SYMPTOM, negated=False



word2vec Embeddings

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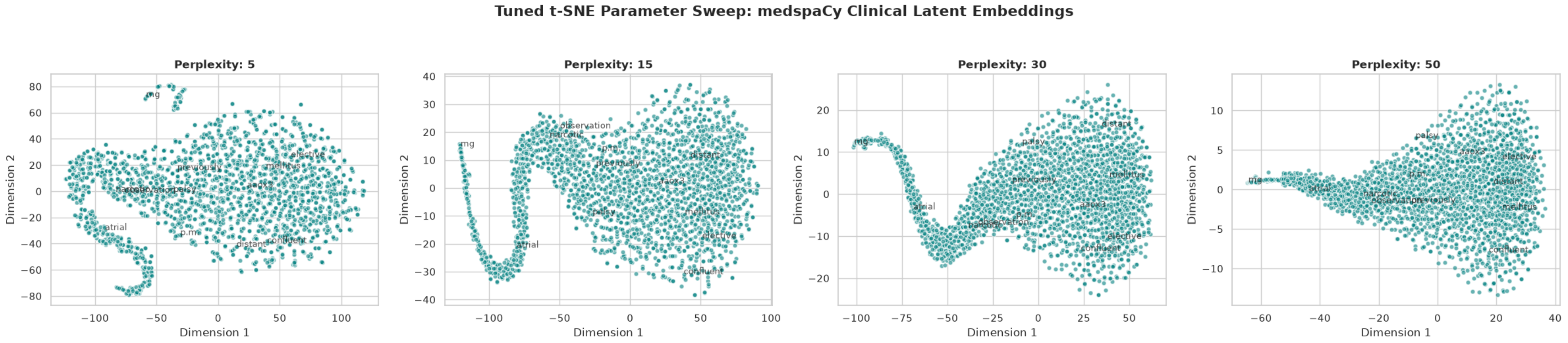
medspaCy — Tuned t-SNE Plot

word2vec embeddings projected to 2D across four perplexity values



Built from the same token vocabulary as the entity pipeline above, independent of how many entities the target rules matched.

RESULTS

Side-by-Side Comparison

	spaCy	scispaCy	medspaCy
Total entities found	3,104	1,986	254
Entity focus	General (people, orgs, dates)	Biomedical (disease, chemical)	Clinical context (negation, etc.)
Top categories found	CARDINAL, ORG, PERSON	DISEASE, CHEMICAL	CONDITION, SYMPTOM
Needs custom setup?	No	No	Yes - target rules required
Negation detection	Not supported	Not supported	58 negations found
word2vec vocab size	2,306	2,455	2,386
Best for	Quick, general text	Clinical concept extraction	Negation & context accuracy

From Entities to Clinical Insight

Key Takeaways

- General NLP misses most clinical meaning in notes
- Domain training (scispaCy) surfaces real clinical concepts
- Context detection (medspaCy) tells a positive finding from a negative one
- Picking the right tool changes what you can reliably extract from notes

How to Reproduce This Analysis

- Get PhysioNet access to MIMIC-III
- Load MIMIC-III into Google BigQuery
- Run the cohort query in the notebook
- Install spaCy, scispaCy, and medspaCy
- Run entity extraction, word2vec, and t-SNE for each tool